Paper Review • arXiv 2611.16728

EgoX

Egocentric Cross-Embodiment Manipulation: Sharing Human, Real-Robot and Sim Experience via "Embodiment Dreaming"

EgoX: Egocentric Cross-Embodiment Manipulation with Embodiment Dreaming

TL;DR Heterogeneous demos are split into semantically aligned modalities in a shared egocentric frame; pretraining with modality-level modules + a shared Transformer backbone + training-only future-latent prediction lifts mean task progress on 13 task–embodiment pairs from 0.63 to 0.80.

7
embodiments (human + 6 robots)

5
manipulation tasks

13
task–embodiment pairs

0.63→0.80
avg. task progress

Yaru Niu, Binghong Chen et al. • CMU / Bosch Center for AI / NVIDIA / UC Berkeley / UT Arlington • first submitted 2026-11-12

Background & Problem

Embodiments are complementary but differ in body, sensing & control

Different morphologies

Humanoid, quadruped, wheeled bimanual, tabletop arm; 5-finger hand, 4-finger hand, gripper, fixed tool

Different sensing

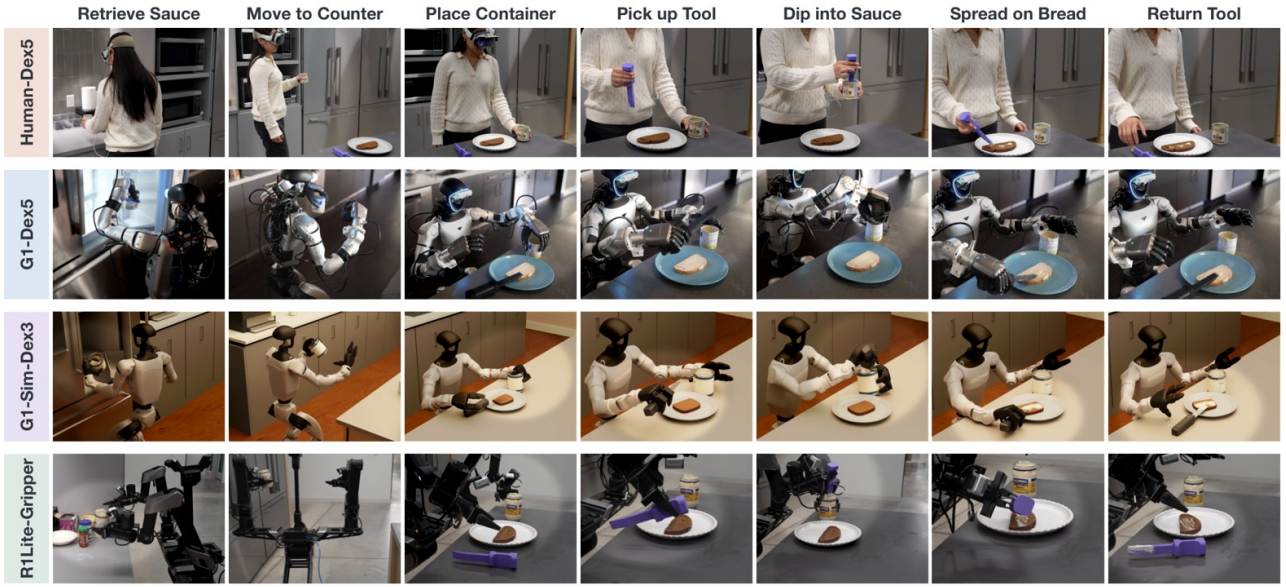
Camera placement and viewpoint vary; proprioception (joint states) differs in dimension and meaning

Different control & dynamics

Action semantics, dimensions and motion dynamics vary; naive mixing rarely yields positive transfer

Key question

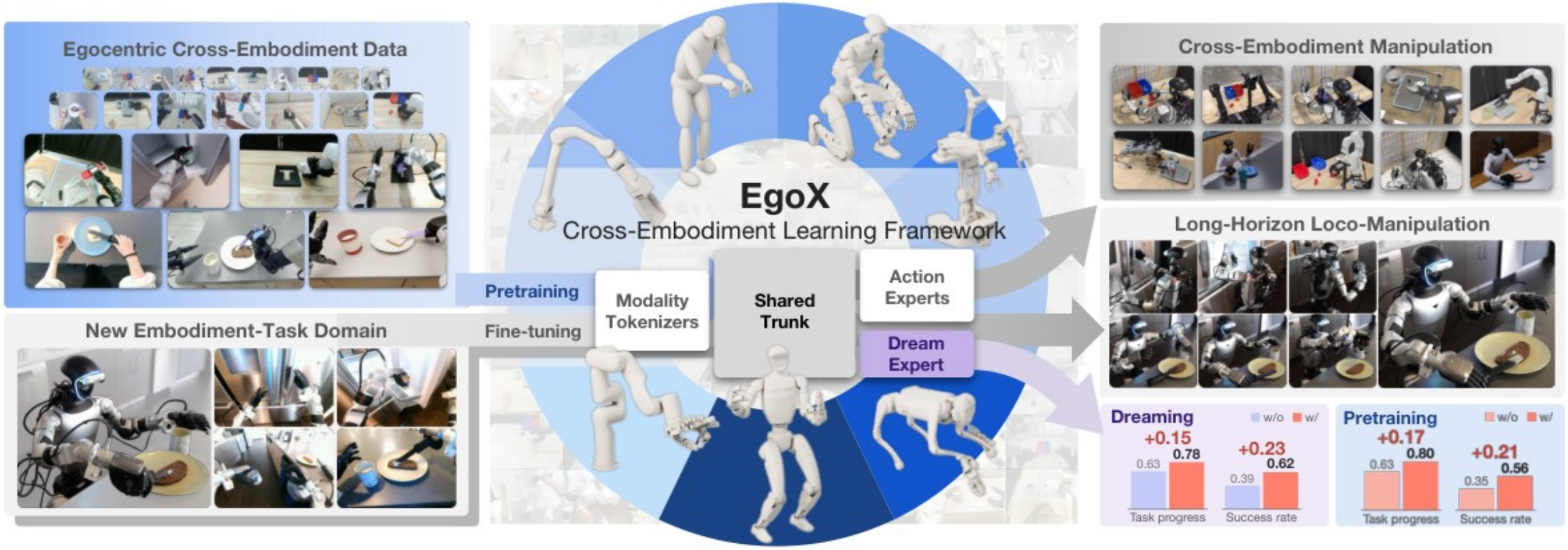
How can we learn shared manipulation knowledge across embodiments while preserving each one's sensing and control differences?



Matching stages of Kitchen-Bread: Human-Dex5, G1-Dex5, G1-Sim-Dex3, R1Lite-Gripper

Method Overview

EgoX: share aligned modalities, keep embodiment-specific interfaces



① Unified modality representation

Human / real / sim data split into image, wrist pose, fingertip keypoints, torso, etc., in one egocentric frame

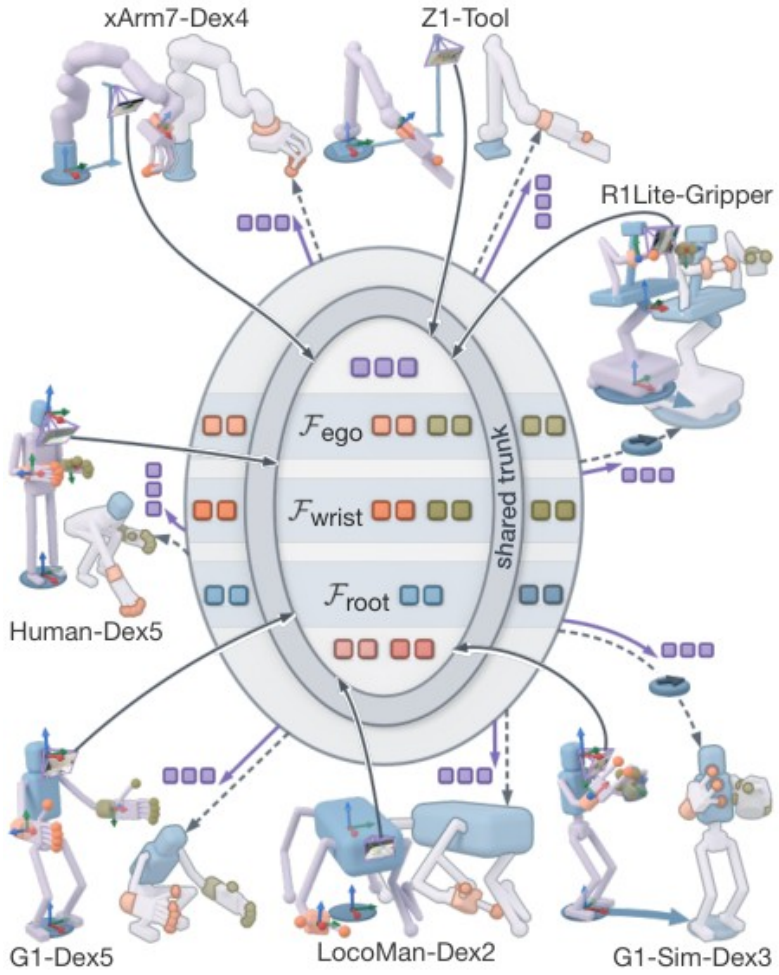
② MXT+ modular model

One tokenizer per modality, one expert per action modality, with a shared Transformer backbone in between

③ Embodiment dreaming

Predicts future visual and proprioceptive latents in training so the backbone learns each embodiment's dynamics

3 egocentric frames + fingertip keypoints make bodies comparable



F_root: torso pose, velocity commands

Ground projection of a reference body part (humanoid feet / quadruped torso / initial head pose for humans)

F_ego: left / right wrist poses & wrist targets

Rigidly attached to the body carrying the main camera; arm base for fixed-base arms

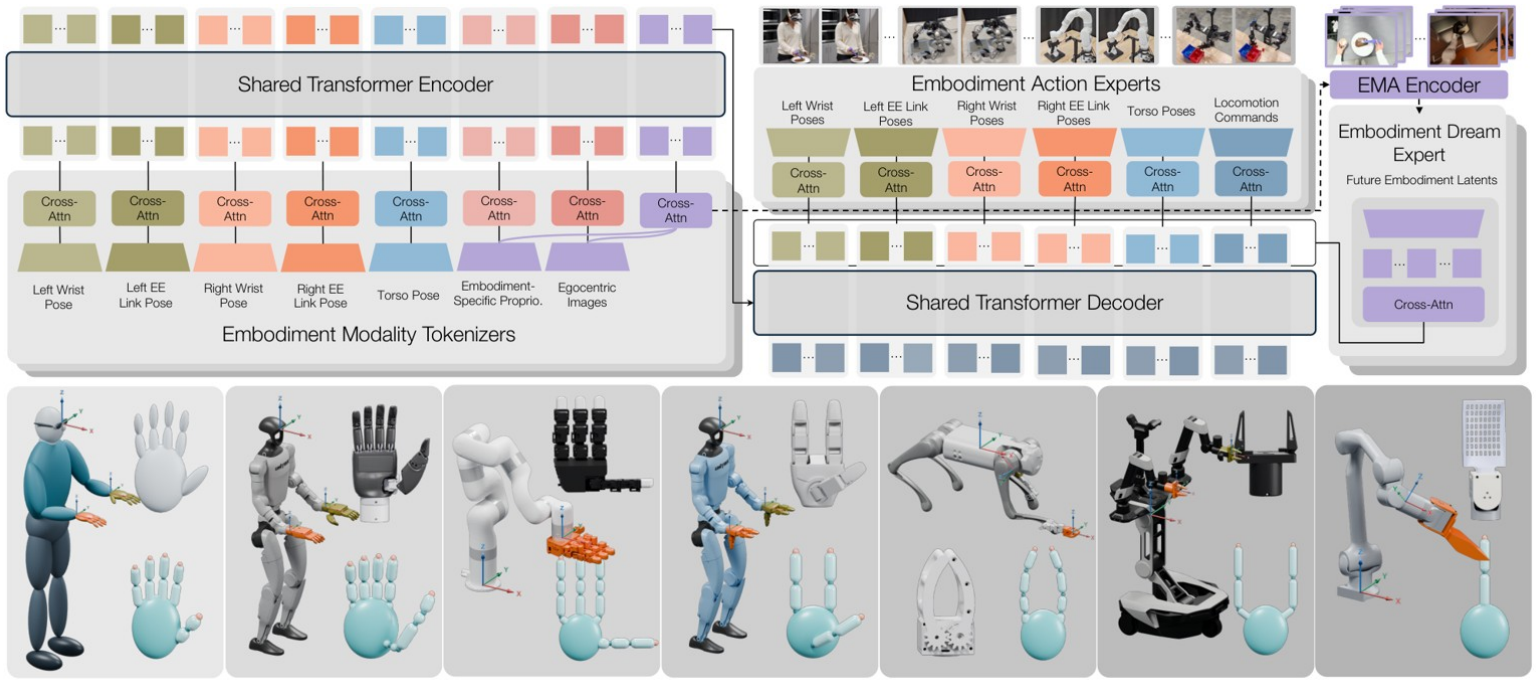
F_wrist: end-effector keypoints

3D positions of fingertips or contact points in the wrist frame; up to 5 slots, masked if missing

Why keypoints rather than joint angles?

5-finger hands, 3-finger hands, grippers and spoons differ entirely in kinematics, yet "where the fingertips are" is directly comparable, so human-hand data can supervise robots. Missing observation modalities are omitted; missing action modalities get no loss.

Modality tokenizers & action experts; a shared backbone transfers



$$\pi^e = \Psi^e \circ G_\theta \circ \Phi^e$$

Φ^e Modality tokenizer

One per embodiment per observation modality; images use a shared ResNet-18 + embodiment-specific aggregator

G_θ Shared backbone

Encoder-decoder Transformer trained by all embodiments; the main channel of transfer

Ψ^e Action expert

One per action modality; outputs 60-step action chunks (2 s at 30 Hz)

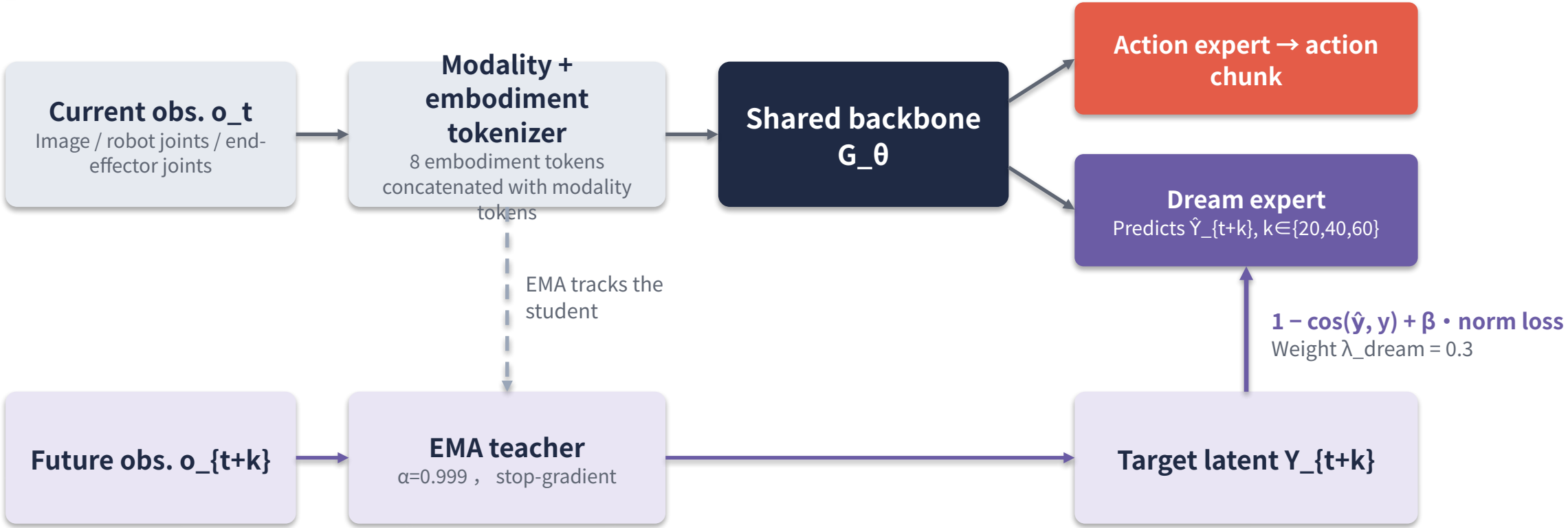
Model size

Pretraining 120M–185M; fine-tuning 50M–90M params

Source: Fig. 3, Appendix A.2

Component ③ Embodiment Dreaming

Predict future embodiment latents in training; no inference cost



Training only
Dream expert & EMA teacher removed at deployment; no extra inference compute

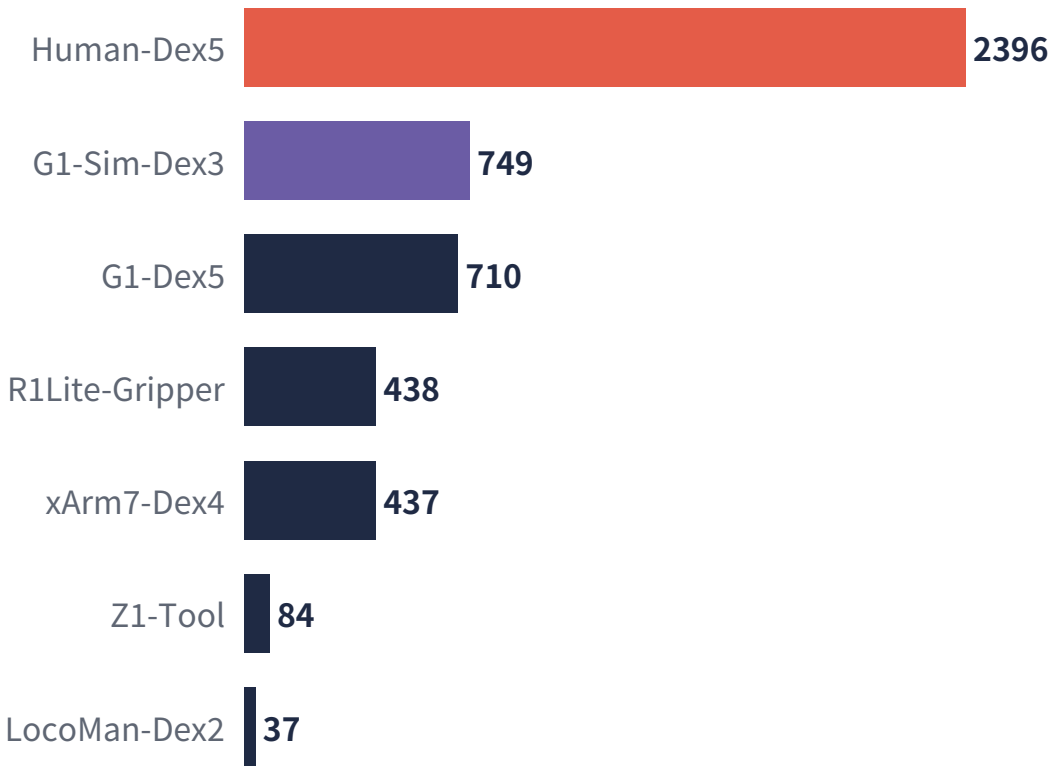
Latent space, not pixels
JEPA-style objective; no high-dimensional image reconstruction

Learns embodiment dynamics
Forces the backbone to encode how each embodiment's visuals and body state evolve

Experimental Setup

7 embodiments, 5 tasks, 13 pairs; human data is nearly half

Demos per embodiment (coral = human, purple = sim, navy = real)



4,851 demos / 34.38 h in total; human 49.4%, real robot 35.2%, sim 15.4%, highly imbalanced across embodiments

5 tasks

Kitchen-Bread	Long-horizon loco-manip.: get sauce from fridge, walk to counter, dip & spread on bread
Kitchen-Cleanup	Sim: put the spatula into a container
Sort	Sort blocks by color into matching bins
Insert	Grasp, align and precisely insert a T-shaped block
Scoop	Scoop cat litter with a spoon and dump it in a trash bin (tool use)

Evaluation protocol

Real robot: 20 trials per pair (13 ID + 7 OOD); sim: 50 (32 + 18). Metric: stage-wise normalized task progress, +1 bonus for success

Note what OOD means

OOD is defined relative to the target embodiment's fine-tuning data; some conditions may appear in other embodiments' (esp. human) pretraining data, so it tests cross-embodiment transfer

Cross-emb. pretraining: avg. progress 0.63 → 0.80, esp. OOD

0.63 → 0.80

Avg. task progress over 13 pairs (MXT+ no pretraining → pretrained)

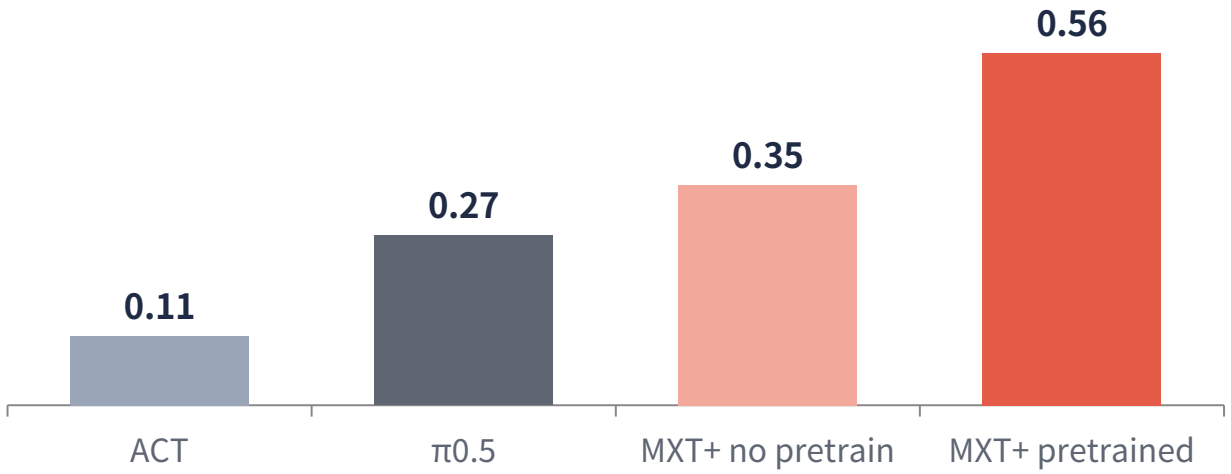
0.53 → 0.77

OOD avg. task progress: improved on 13/13 pairs (ID: 11/13)

12 / 13

Pairs where pretrained MXT+ ranks first among all methods

Avg. success rate over 13 pairs (Appendix A.10)



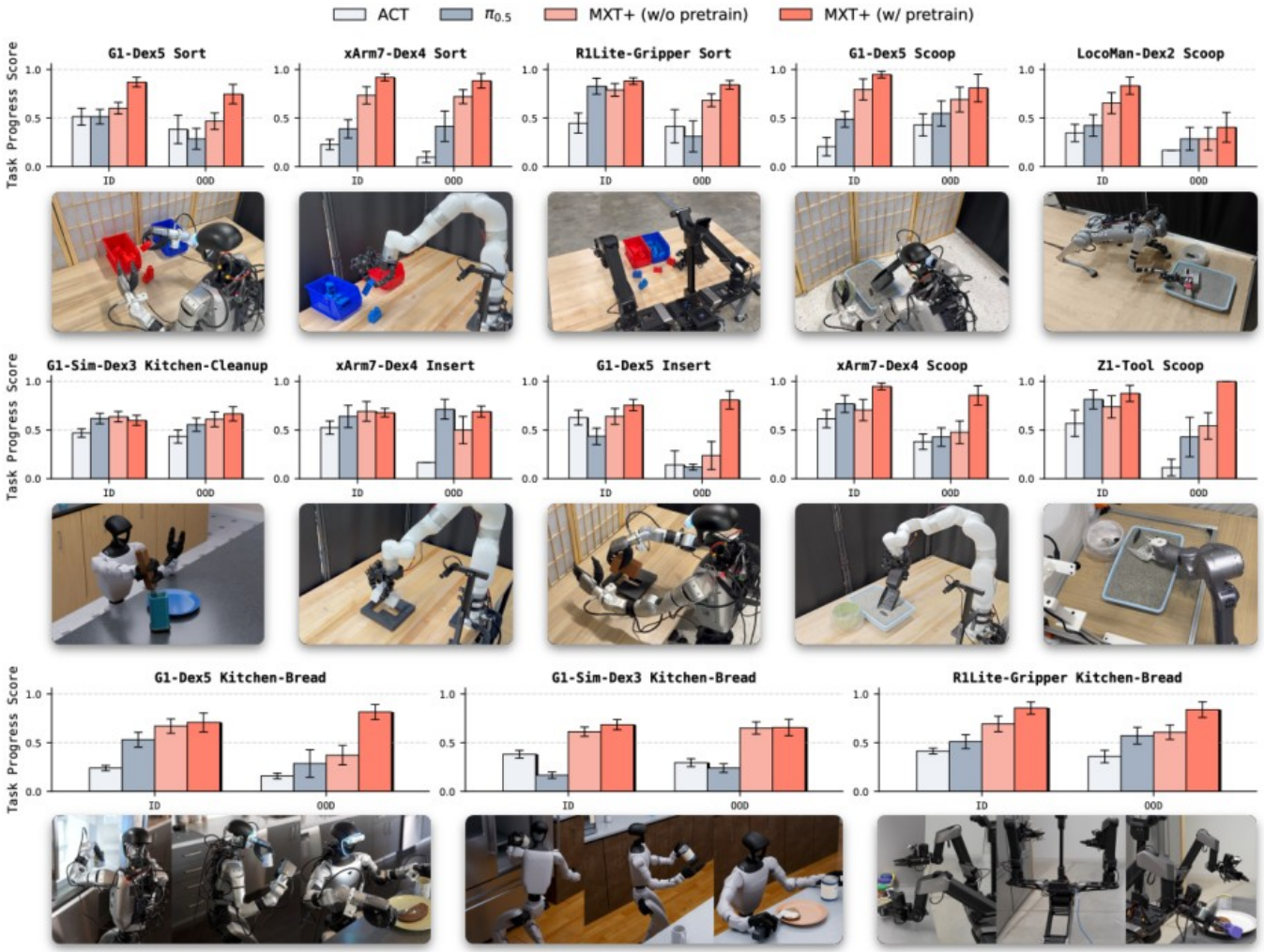
OOD success: pretraining opens the gap

MXT+ w/o pretraining 0.18 ≈ π0.5 0.19 → pretrained MXT+ 0.52; ID: 0.44 → 0.58

Small models are competitive

MXT+ fine-tunes at 50M–90M vs. ~3.4B for π0.5; even without pretraining, its avg. success rate beats π0.5 (0.35 vs 0.27)

Largest OOD gains: G1-Dex5 Insert and Z1-Tool Scoop



G1-Dex5 Insert • OOD

No pretraining ≈0.23 → pretrained ≈0.82

Z1-Tool Scoop • OOD

No pretraining ≈0.55 → pretrained 1.0

Exceptions & ties

xArm7-Dex4 Insert: π0.5 close to pretrained MXT+ (OOD ≈0.72 vs 0.70); MXT+ has a lower success rate

G1-Sim Kitchen-Cleanup: all methods around 0.6

Note: values marked "≈" are approximate readings from figures

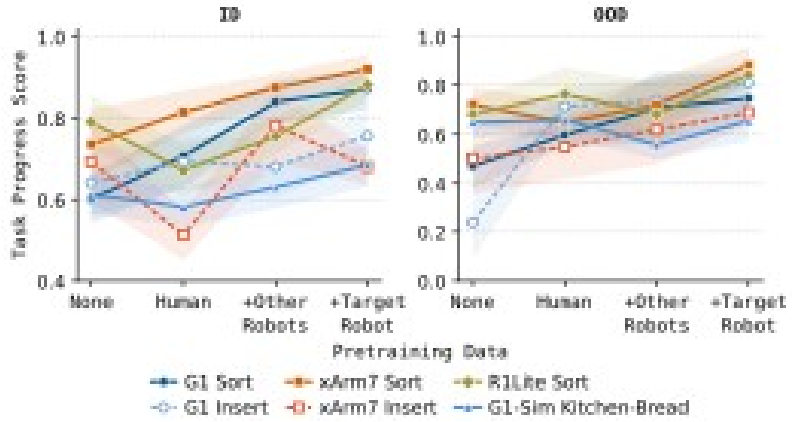
Broader mix → better transfer; human data gives transferable priors

No pretraining
Target-embodiment, target-task data only

+ Human demos
OOD gains on 5/6 pairs, most for G1-Dex5 Insert

+ Other robots
ID gains on 5/6 pairs

+ Target robot's other tasks
OOD gains on 6/6 pairs: full cross-embodiment pretraining

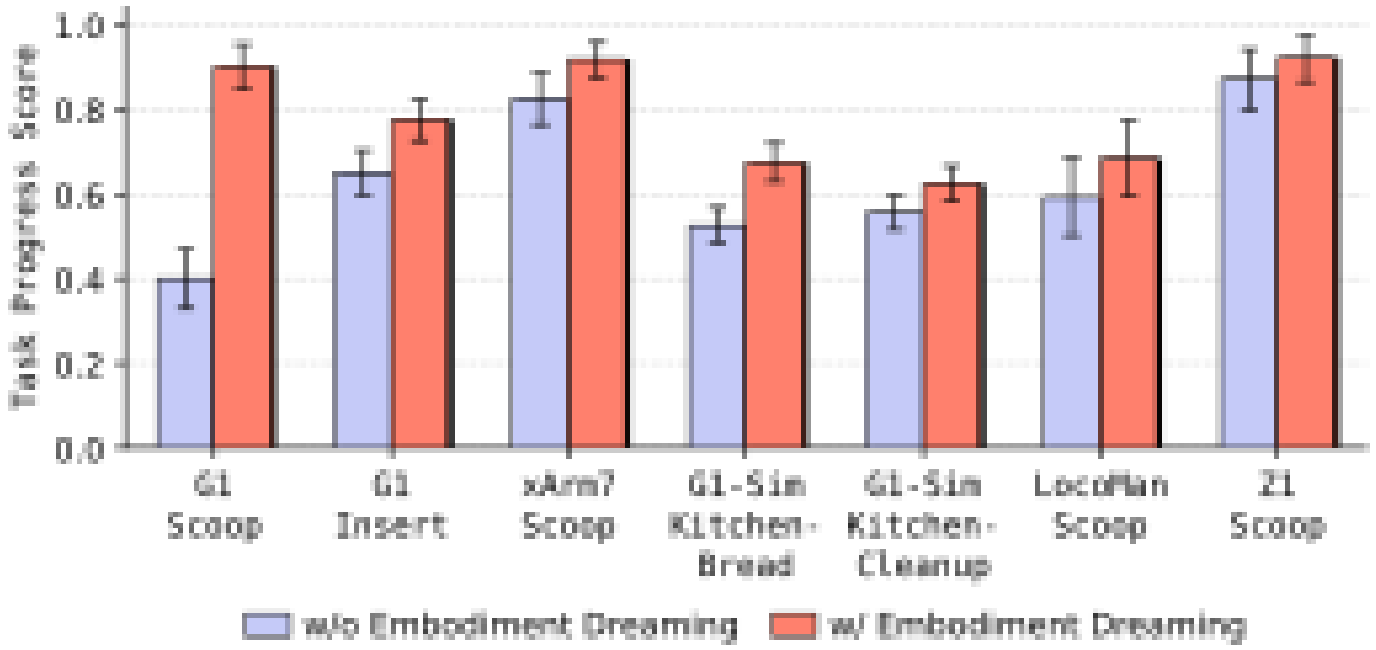


+0.12
ID: full mix vs. no pretraining

+0.23
OOD: full mix vs. no pretraining

Source: Fig. 7 left (Sec. 4.2)

Without dreaming, performance drops broadly; all 7 settings benefit



Task progress of MXT (no dreaming) vs MXT+ (dreaming) on 7 task–embodiment settings

0.63 → 0.78
Avg. task progress (all 7 settings improve)

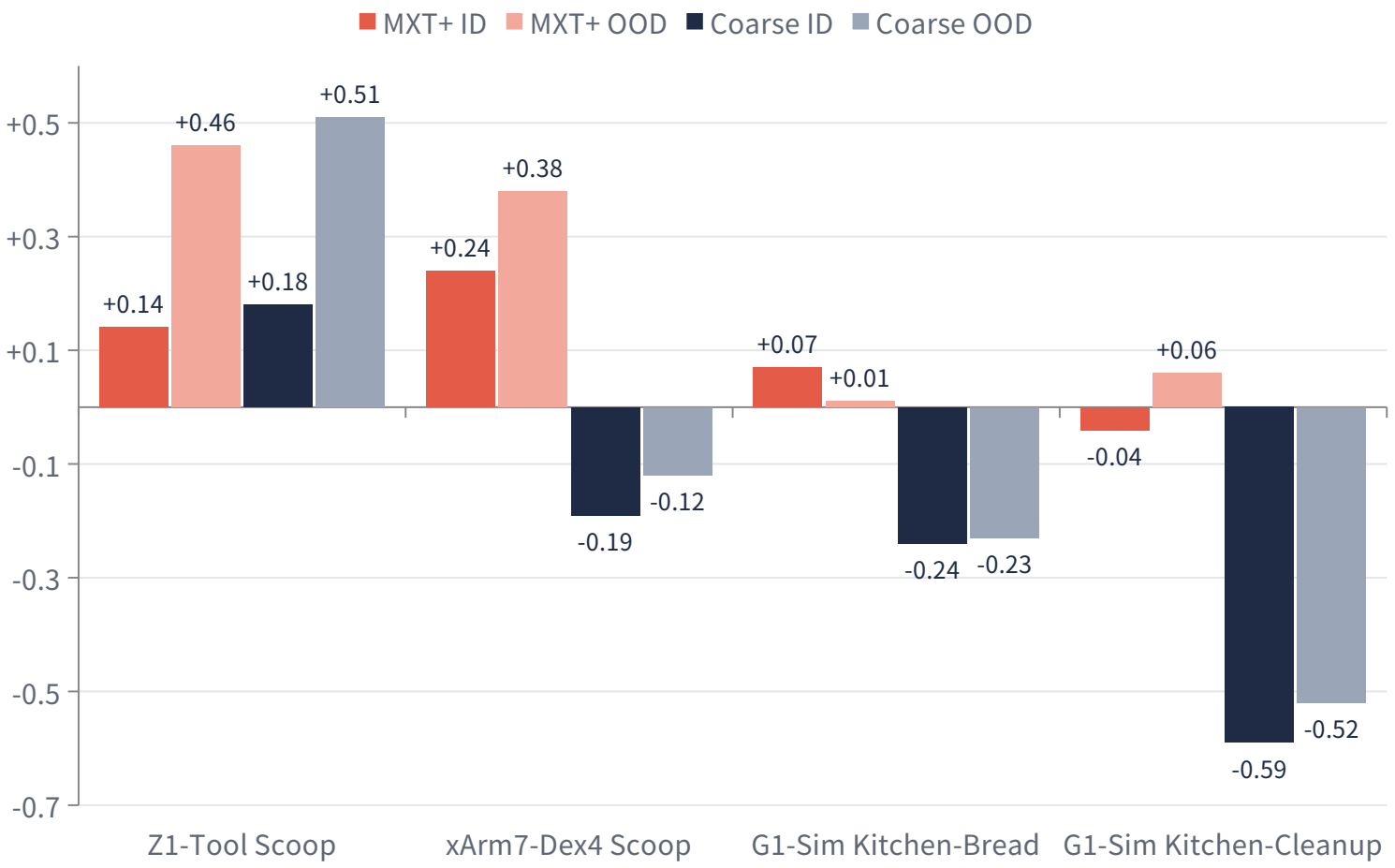
0.39 → 0.62
Avg. success rate

0.27 → 0.60
OOD avg. success rate (ID: 0.46 → 0.63)

3 → 16 / 20
G1-Dex5 Scoop successes (largest gain)

Rich modalities derail coarse sharing; modularity avoids big drops

Task progress change Δ from cross-embodiment pretraining (none → pretrained)



Coarse variant

1 tokenizer for all cameras; 1 for concatenated pose + proprio; 1 expert for all actions

Richer modalities, larger gap

Z1-Tool has few modalities, so both benefit; coarse pretraining degrades xArm7 and sim G1

Kitchen-Cleanup drops to zero

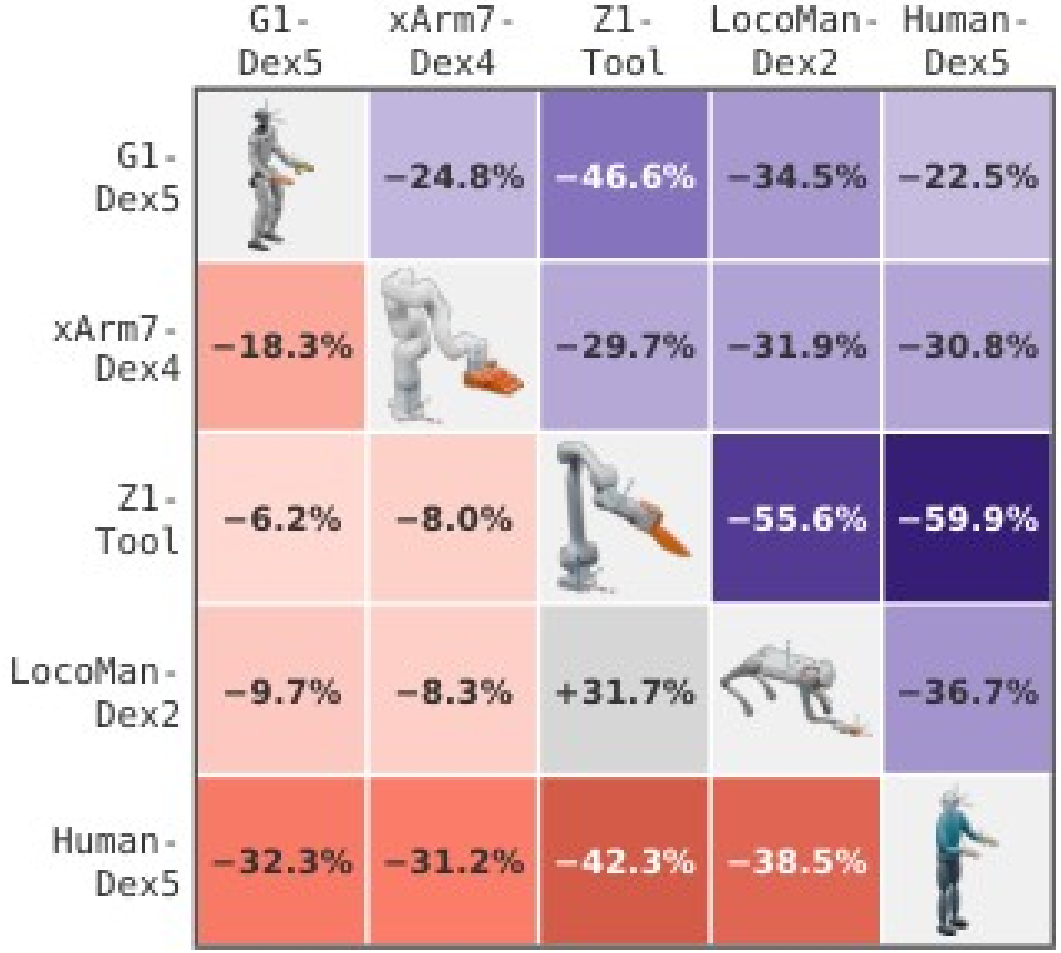
Episodes start with walking; the 3-D velocity command is only 3 of 51 action dims and may be drowned out by coarse sharing

After pretraining

Modular MXT+ is never worse than coarse in any ID / OOD setting

Source: Fig. 8 (Δ values annotated in figure)

Pretraining & dreaming align backbone geometry across embodiments



GW distance change: lower triangle = pretraining (coral), upper = dreaming (purple)

−21.30%

Pretraining: avg. GW (drops for 9/10 pairs)

−38.52%

Dreaming: avg. GW (drops for 10/10 pairs)

How it is measured

On Scoop demos, take the right-wrist tokens from the shared backbone output, mean-pool to 512-D, and compute pairwise Gromov–Wasserstein distances between embodiments; lower = more similar geometry

Details

With pretraining, human–robot pairs drop most (−31% to −42%); the only increase is LocoMan–Z1 (+31.7%)

With dreaming: Z1–Human −59.9%, Z1–LocoMan −55.6%

Assessment

Systematic, wide real-robot coverage; scale & rigor still need work

Strengths

Very broad embodiment coverage

Humanoid, wheeled bimanual, quadruped, tabletop arm, sim + human; 5-finger hands to grippers to fixed tools, all with real-robot evaluation

Simple, effective representation

Fingertip keypoints + egocentric frames sidestep incomparable joint spaces; human-hand data supervises directly

Dreaming is cost-effective

Training-only, zero inference cost, gains on 7/7 settings, backed by latent-geometry evidence

Instructive modularity ablation

Reveals the negative-transfer risk of coarse sharing when modalities are rich

Limitations & Questions

Small data and trial scale

~34 h, 4.8k trajectories, all self-collected; 20 real-robot trials per pair, only 7 OOD

Loose OOD definition

Some OOD conditions appear in other embodiments' pretraining data, closer to "cross-embodiment transfer" than generalization

Comparisons not fully matched

π0.5 only LoRA-fine-tuned on target data; no comparison with other cross-embodiment architectures on the same data

Scaling & sampling cost

New embodiments need manual modality mapping & frame transforms; balanced sampling reuses small datasets heavily

Summary & Takeaways

01 Share semantically aligned modalities; keep per-embodiment interfaces

Unified representation + modality-level tokenizers / experts + a shared backbone enable positive transfer among human, real-robot and sim data in one model

02 Cross-embodiment pretraining pays off mainly OOD

OOD task progress 0.53 → 0.77, success rate 0.18 → 0.52; broader pretraining mixes are better, and human data helps too

03 Predicting "the body's future" is a cheap, effective auxiliary signal

Embodiment dreaming: task progress 0.63 → 0.78, zero inference cost, more similar cross-embodiment latent geometry (GW −38.52%)

Next to verify: larger, more balanced data; automated embodiment onboarding; and comparisons with other cross-embodiment architectures on equal data.